

**Sri Chaitanya** IIT Academy.,India.

A.P. T.S. KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S) & LIIT

JEE-MAIN

Date: 28-12-2025

Time: 09.00Am to 12.00Pm

GTM-06

Max. Marks: 300

KEY SHEET**MATHEMATICS**

1	3	2	3	3	3	4	3	5	3
6	2	7	3	8	2	9	1	10	3
11	4	12	2	13	2	14	3	15	2
16	1	17	3	18	2	19	4	20	2
21	182	22	5	23	1	24	12	25	4

PHYSICS

26	2	27	3	28	2	29	1	30	2
31	2	32	2	33	4	34	4	35	4
36	1	37	4	38	4	39	1	40	3
41	4	42	3	43	3	44	4	45	4
46	6	47	5	48	2	49	400	50	10

CHEMISTRY

51	2	52	3	53	2	54	2	55	2
56	1	57	3	58	3	59	3	60	3
61	3	62	4	63	2	64	2	65	3
66	4	67	4	68	2	69	2	70	2
71	3	72	6	73	5	74	2	75	2



SOLUTION MATHEMATICS

1. $\lim_{x \rightarrow 0} \frac{f(x^2 + x + 1)}{f(x^4 - x^2 + 2x + 4) - f(4)} \left(\frac{0}{0} \text{ form} \right)$

As. $f(x)$ is differentiable then apply L'Hospital rule

$$\lim_{x \rightarrow 0} \frac{(2x+1)f'(x^2+x+1)}{(4x^3-2x+2)f'(x^4-x^2+2x+4)} = \frac{f'(1)}{2f'(4)} = 4$$

2. Let $h = 2t_1^2$ & $12 = 4t_2$

We know $t_2 = -t_1 - \frac{2}{t_1} \Rightarrow t_1 = -1$ or $-2 \Rightarrow h = 2$ or 8

3. Focus is foot of perpendicular drawn from vertex on latus rectum i.e. $(4, 3)$ and equation of the directrix is $x + y + c = 0$

i.e. $\left| \frac{1+c}{\sqrt{2}} \right| = 3\sqrt{2} \Rightarrow c = 5$ or -7

$\therefore$ equation of the directrix is $x + y + 5 = 0$

$\therefore$ equation of the required parabola is

$$(x-4)^2 + (y-3)^2 = \left(\frac{x+y+5}{\sqrt{2}} \right)^2$$

$$\Rightarrow x^2 + y^2 - 2xy - 26x - 22y + 25 = 0$$

Compare with $x^2 + y^2 - 2xy + px + qy + r = 0$

$$\Rightarrow p = -26, \quad q = -22, \quad r = 25 \quad \therefore p + q + r = -23$$

4. $y^2 = 2cx + 2c^{3/2} \dots\dots\dots(1)$

$$2y \frac{dy}{dx} = 2c \quad y \frac{dy}{dx} = c \quad \dots\dots\dots(2)$$

from (1) and (2)

$$y^2 = 2y \frac{dy}{dx} (x) + 2 \left(\frac{ydy}{dx} \right)^{3/2} \quad \left(y - 2x \frac{dy}{dx} \right) = 2 \left(\frac{dy}{dx} \right)^{3/2} y^{1/2}$$

$$\left(y - 2x \frac{dy}{dx} \right) = 2 \left(\frac{dy}{dx} \right)^{3/2} y^{1/2}$$

on squaring

$$\text{degree} = 3 \quad \text{order} = 1$$

5. **Statement-1:** General term $= {}^{1024}C_r (5^{1/2})^{1024-r} ((7)^{1/8})^r$ will be

integer. if r is multiple of 8

$$\Rightarrow r = 0, 8, 16, 24, \dots\dots\dots, 1024$$



$\Rightarrow$ Number of integral terms = 129.

Statement-2: General term = $\frac{10!}{\alpha!\beta!\gamma!} 2^{\alpha/2} 3^{\beta/3} 5^{T/6}$ for rational terms

$$\alpha = 0, 2, 4, 6, 8, 10 \quad \beta = 0, 3, 6 \quad \gamma = 0, 6$$

Hence possible sets = $(4, 6, 0), (4, 0, 6); (10, 0, 0)$

Hence, there are 3 rotational terms.

$$\therefore \text{required rdom} = \frac{10!}{4!6!} 2^2 5 \frac{10!}{10!} 2^5 = 12632.$$

Statement-3: t_{r+1} , the $(r+1)$ in the expansion of $(5^{1/6} + 2^{1/8})^{10}$ is

$$t_{r+1} = {}^{100}C_r (5^{1/6})^{100-r} (2^{1/8})^r$$

As 5 and 2 are relatively prime, t_{r+1} will be rational if $\frac{100-r}{6}$ and $\frac{r}{8}$ are both integers.

i.e if $100-r$ is a multiple of 6 and r is a multiple of 8. As $0 \leq r \leq 100$, multiple of 8 upto 100 and corresponding value of $100-r$

$$r = 0, 8, 16, 24, \dots, 88, 96$$

$$100-r = 100, 92, 84, 76, \dots, 12, 4$$

Out of $100-r$, multiple of 6 are 84, 60, 36, 12

$\therefore$ there are just four rational terms

$\Rightarrow$ number of irrational terms is $101-4=97$

$$6. \quad g(x) = x^2 + \frac{1}{x^2} \quad f(g(x)) = x^6 + \frac{1}{x^6} = \left(x^2 + \frac{1}{x^2}\right)^3 - 3\left(x^2 + \frac{1}{x^2}\right)$$

$$f(g(x)) = (g(x))^3 - 3g(x) \Rightarrow f(x) = x^3 - 3x \quad f'(x) = 3x^2 - 3$$

$$7. \quad f(x) = |\cos x| + \cos^{-1}(\operatorname{sgn} x) + |\ln x|$$

$$f(x) = |\cos x| + \cos^{-1}(1) + |\ln x| \quad \forall x \in (0, 2\pi)$$

$$= |\cos x| + |\ln x| \quad \forall x \in (0, 2\pi) \quad \therefore f(x) \text{ is not differentiable at } x = 1, \frac{\pi}{2}, \frac{3\pi}{2}$$

8. Let middle point of chord be (h, k)

$$T = S_1 \Rightarrow \frac{xh}{3} + \frac{yk}{4} = \frac{h^2}{3} + \frac{k^2}{4}$$

$$\text{Chord passing through } (2, 0) \Rightarrow \frac{2h}{3} = \frac{h^2}{3} + \frac{k^2}{4}$$

$$\Rightarrow \frac{(h-1)^2}{1} + \frac{k^2}{\frac{4}{3}} = 1 \Rightarrow \frac{(x-1)^2}{1} + \frac{y^2}{\frac{4}{3}} = 1$$

$$9. \quad \underbrace{A, I, A, I, O}_5 \text{ \& } \underbrace{P, P, L, C, T, N}_6$$



$$\text{Ways} = \frac{6!}{2!} \cdot {}^7C_5 \cdot \frac{5!}{2!2!} = (45)7!$$

10. Let $x_i - 5 = d_i$

$$\sigma_x^2 = \sigma_d^2 = \frac{\sum d_i^2}{n} - \left(\frac{\sum d_i}{n} \right)^2 = \frac{125}{10} - \left(\frac{5}{10} \right)^2 = \frac{25}{2} - \frac{1}{4} = \frac{49}{4}$$

11. Total ways $\Rightarrow \frac{4.6.5.4.(\alpha+1)}{2}$

factor having HCF 1 $\Rightarrow 2^{5-1} = 16$

factor having HCF 5 $\Rightarrow 2^{4-1} \cdot 2 = 16$

factor having HCF 7 $\Rightarrow 2^{4-1} \cdot 2 = 16$

Probability $\Rightarrow \frac{48.2}{4.6.5.4(\alpha+1)} = \frac{1}{35} \Rightarrow \alpha = 6$

12. Here $|A| = 4$ where $A = \begin{bmatrix} x & 1 & 1 \\ 2 & x & 3 \\ x & -1 & 2 \end{bmatrix}$, then $\begin{vmatrix} 2x+3 & -3 & 3-x \\ 3x-4 & x & 2-3x \\ -x^2-2 & 2x & x^2-2 \end{vmatrix} = |adj A| = |A|^2 = 16$

13. We know that $e^x = \sum_{r=0}^{\infty} \frac{x^r}{r!}$, let $x = e^{i\theta} \cdot \cos \theta$

Let $e^{\cos \theta \cdot e^{i\theta}} = \sum_{r=0}^{\infty} \left(\frac{\cos^r \theta \cdot e^{i(r\theta)}}{r!} \right) \Rightarrow e^{\cos \theta (\cos \theta + i \sin \theta)} = \sum_{r=0}^{\infty} \frac{\cos^r \theta (\cos r\theta + i \sin r\theta)}{r!}$

$$\Rightarrow \sum_{r=0}^{\infty} \frac{\cos r\theta \cdot \cos^r \theta}{r!} + i \sum_{r=0}^{\infty} \frac{\cos^r \theta \sin r\theta}{r!} = e^{\cos^2 \theta} \cdot e^{i(\sin \theta \cos \theta)}$$

$$\Rightarrow x_2 = e^{\cos^2 \theta} \cos(\sin \theta \cos \theta) \Rightarrow x_1 = e^{\cos^2 \theta} \sin(\sin \theta \cos \theta)$$

$$\Rightarrow x_1^2 + x_2^2 = e^{2\cos^2 \theta}; 2x_1x_2 = e^{2\cos^2 \theta} \sin(2\sin \theta \cos \theta)$$

14. The plane containing the given line is

$$(2x + 3y + 5z + 1) + \lambda(3x + 4y + 6z + 2) = 0$$

$$\Rightarrow (2 + 3\lambda)x + (3 + 4\lambda)y + (5 + 6\lambda)z + (1 + 2\lambda) = 0$$

This plane is parallel to y - axis

$$\Rightarrow 3 + 4\lambda = 0 \Rightarrow \lambda = -\frac{3}{4}$$

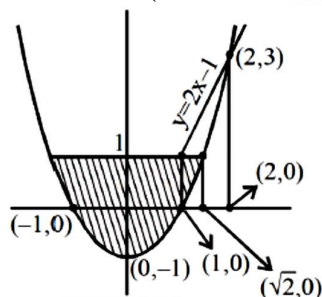
$\Rightarrow$ A point on y - axis is the origin.

$\Rightarrow$ Perpendicular distance of the origin from the

Plane $x - 2z + 2 = 0$ is $\frac{2}{\sqrt{5}}$



$$\begin{aligned}
 15. \quad S &= \frac{1}{1^5} + \frac{1}{3^5} + \frac{1}{5^5} + \frac{1}{7^5} + \dots \\
 &= \left(\frac{1}{1^5} + \frac{1}{2^5} + \frac{1}{3^5} + \frac{1}{4^5} + \dots \right) - \left(\frac{1}{2^5} + \frac{1}{4^5} + \frac{1}{6^5} + \dots \right) \\
 &= (k) - \frac{1}{32} \left(\frac{1}{1^5} + \frac{1}{2^5} + \frac{1}{3^5} + \dots \right) = k - \frac{k}{32} = \frac{31k}{32}
 \end{aligned}$$



$$16. \quad y = \begin{cases} x-1+x; & x > 1 \\ -x+1+x; & x \leq 1 \end{cases} = \begin{cases} 2x-1; & x > 1 \\ 1; & x \leq 1 \end{cases}$$

Required area

$$= \int_{x=-\sqrt{2}}^{x=\sqrt{2}} [1 - (x^2 - 1)] dx + \int_{y=1}^{y=3} \left[\sqrt{y+1} - \frac{y+1}{2} \right] dy = \frac{7+4\sqrt{2}}{3}$$

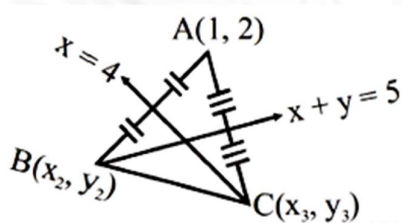
$$17. \quad S_1: (x-1)^2 + (y-1)^2 + \lambda(x+y-2) = 0$$

$$S_2: x^2 + y^2 + 4x + 5y - 4 = 0$$

$$\text{chord } PQ: S_1 - S_2 = 0$$

$$(-6x - 7y + 6) + \lambda(x - y - 2) = 0$$

$$\Rightarrow x = 8, y = -6$$



18.

$$\frac{x_2 + 1}{2} = 4 \Rightarrow x_2 = 7$$

$$x_2 + y_2 = 5 \Rightarrow 7 + y_2 = 5$$

$$\Rightarrow y_2 = -2 \quad \therefore (x_2, y_2) \equiv (7, -2)$$

$$x_3 = 4$$

$$\frac{x_3 + 1}{2} + \frac{y_3 + 2}{2} = 5 \Rightarrow 4 + y_3 = 10 - 3 = 7$$

$$\Rightarrow y_3 = 3 \quad \therefore (x_3, y_3) \equiv (4, 3)$$

$$19. \quad \text{Statement-1: } |adj(adjA)| = |A|^{(n-1)^2}$$



$$|adj(adj(adj(A)))| = |A|^{(n-1)^3} = 3^8$$

Statement-2: $|A^{-1} \cdot adj(B^{-1}) \cdot adj(2A^{-1})| = \frac{1}{|A|} \cdot \frac{1}{|B|^2} \cdot |2A^{-1}|^2$

$$= \frac{1}{|A|} \cdot \frac{1}{|B|^2} \cdot \frac{64}{|A|^2} = \frac{64}{|A|^3 |B|^2} = \frac{64}{8 \times 9} = \frac{8}{9}$$

Statement-3:

$$3ABA^{-1} + 3A = 2A^{-1}BA + 2A$$

$$3ABA^{-1} + 3AAA^{-1} = 2A^{-1}BA + 2A^{-1}AA$$

$$|3A(B+A)A^{-1}| = |2A^{-1}(B+A)A|$$

$$3^n |A+B| = 2^n |A+B|$$

$$\Rightarrow |A+B| = 0$$

20. Let $I = \int_0^{\sin^2 x} \sin^{-1}(\sqrt{t}) dt + \int_0^{\cos^2 x} \cos^{-1}(\sqrt{t}) dt$

$$= \left[t \sin^{-1}(\sqrt{t}) \right]_0^{\sin^2 x} - \int_0^{\sin^2 x} \frac{\sqrt{t}}{2\sqrt{1-t}} dt + \left[t \cos^{-1}(\sqrt{t}) \right]_0^{\cos^2 x} + \int_0^{\cos^2 x} \frac{\sqrt{t}}{2\sqrt{1-t}} dt$$

$$= x \sin^2 x + \int_{\sin^2 x}^0 \frac{\sqrt{t}}{2\sqrt{1-t}} dt + x \cos^2 x + \int_0^{\cos^2 x} \frac{\sqrt{t}}{2\sqrt{1-t}} dt$$

$$= x(\sin^2 x + \cos^2 x) + \int_{\sin^2 x}^{\cos^2 x} \frac{\sqrt{t}}{2\sqrt{1-t}} dt$$

Putting $t = \sin^2 \theta$ and $dt = 2 \sin \theta \cos \theta d\theta$, we have

$$\int \frac{\sqrt{t}}{2\sqrt{1-t}} dt = \int \frac{\sin \theta}{\sqrt{1-\sin^2 \theta}} \sin \theta \cos \theta d\theta$$

$$= \int \sin^2 \theta d\theta \int \frac{1-\cos 2\theta}{2} d\theta = \frac{\theta}{2} - \frac{\sin 2\theta}{4}$$

Also, when $t = \sin^2 x$, then $\theta = x$ and when $t = \cos^2 x$, then $\theta = \frac{\pi}{2} - x$.

Hence we have $I = x + \left[\frac{\theta}{2} - \frac{\sin 2\theta}{4} \right]_{(\pi/2)-x}^{(\pi/2)-x}$

$$= x + \left(\frac{\pi}{4} - \frac{x}{2} - \frac{\sin 2x}{4} \right) - \left(\frac{x}{2} - \frac{\sin 2x}{4} \right) = x + \frac{\pi}{4} - x = \frac{\pi}{4}$$

21. The bug starts from A and after 7 steps return to A. So, in its journey we can classify based on number of times it returns to A in intermediate steps.

Case : I (0 times) $R.P = \frac{2^5}{3^6}$

Case : II (1 times) $R.P = 4 \cdot \frac{2^3}{3^5}$

Case : III (1 times) $R.P = 3. \frac{2}{3^4}$

$$\text{So, probability} = \frac{182}{729}$$

22. Let $\vec{c} = \lambda \vec{a} + \mu \vec{b}$ take dot by $\vec{b}$

$$0 = \lambda(\vec{a} \cdot \vec{b}) + (\vec{b})^2 = -\lambda + 5\mu \quad \Rightarrow \quad \lambda - 5\mu = 0 \quad \dots\dots\dots(1)$$

again $\vec{a}.\vec{c}=7 \Rightarrow \lambda \vec{a}^2 + \mu(\vec{a}.\vec{b})=7$

$$\Rightarrow 3\lambda - \mu = 7 \quad \dots\dots\dots(2)$$

solving (1) and (2) $\lambda = \frac{5}{2}, \mu = \frac{1}{2}$

$$\therefore \vec{c} = \frac{5}{2}(-\hat{i} + \hat{j} + \hat{k}) + \frac{1}{2}(2\hat{i} + \hat{k}) = -\frac{3\hat{i}}{2} + \frac{5}{2}\hat{j} + 3\hat{k} \Rightarrow \frac{2}{7}|\vec{c}|^2 = \frac{1}{7} \times 35 = 5$$

23. Equation of chord is $\frac{x-5}{\cos \alpha} = \frac{y-3}{\sin \alpha} = r$

Point $(5 + r \cos \alpha, 3 + r \sin \alpha)$ of chord is one of its end point

$$2(5 + r \cos \alpha)^2 - 3(3 + r \sin \alpha)^2 = 6$$

(5, 3) is midpoint of chord iff sum of roots = 0

$$20\cos\alpha - 18\sin\alpha = 0$$

$$\frac{\cos \alpha}{9} = \frac{\sin \alpha}{10} \Rightarrow \tan \alpha = \frac{10}{9} = m \quad \therefore \quad p - q = 1$$

24. $9\cos^{12}x + \cos^2 2x + 1 = 6\cos^6 x \cos 2x + 6\cos^6 x$

$$\left(3\cos^6 x - 1 - \cos 2x\right)^2 = 0$$

$$3\cos^6 x = 1 + \cos 2x = 2\cos^2 x$$

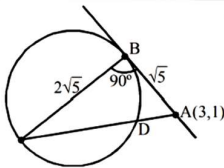
$$\Rightarrow \cos x = 0 \text{ or } \cos^2 x = \frac{2}{3}$$

$$\cos x = 0, \cos x = \pm \sqrt{\frac{2}{3}}$$

$\cos x = 0$ has 4 solutions.

$$\cos x = \pm \sqrt{\frac{2}{3}} \text{ has 8 solutions.}$$

Hence total 12 solutions.



25.

$$\angle ABC = 90^0$$

$$BC = 2\sqrt{5}$$

$$AB = \sqrt{9 + 1 - 5} = \sqrt{5}$$

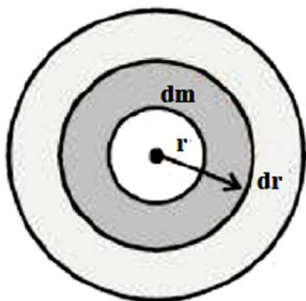
$$\Rightarrow AC = 5$$

Power of point $\Rightarrow AB^2 = AD.AC$

$$\Rightarrow AD = 1 \Rightarrow CD = 4$$

**PHYSICS**

$$\begin{aligned}
 26. \quad dI &= (dm)r^2 \\
 &= (\sigma dA)r^2 \\
 &= \left(\frac{\sigma_0}{r} 2\pi dr \right) r^2 = (\sigma_0 2\pi r dr)
 \end{aligned}$$



$$\begin{aligned}
 I &= \int dI = \int_a^b \sigma_0 2\pi r^2 dr \\
 &= \sigma_0 2\pi \left(\frac{b^3 - a^3}{3} \right)
 \end{aligned}$$

$$\begin{aligned}
 m &= \int dm = \int \sigma dA \\
 &= \sigma_0 2\pi \int_a^b dr
 \end{aligned}$$

$$\begin{aligned}
 27. \quad \frac{1}{\lambda} &= R \left(\frac{1}{m^2} - \frac{1}{n^2} \right) z^2 \\
 \frac{1}{1085} &= R \left(\frac{1}{m^2} - \frac{1}{n^2} \right) 2^2 \\
 \therefore m &= 2 \\
 \therefore n &= 5
 \end{aligned}$$

$$28. \text{ Heat current } H = \frac{-K 2\pi \times l dT}{dx}$$

$$H \int_{r_1}^{r_2} \frac{dx}{x} = -Kl2\pi \int_{T_1}^{T_2} dT$$

$$H \log \left(\frac{r_2}{r_1} \right) = -Kl2\pi (T_2 - T_1)$$

$$H = 2\pi lK (T_2 - T_1) \log \left(\frac{r_2}{r_1} \right)$$



$$\therefore H \propto \log \left(\frac{r_2}{r_1} \right)$$

29. $M_1 = i\pi r^2 N_1$

$$M_2 = ia^2 N_2$$

$$N_1 = 1$$

$$N_2 = 2$$

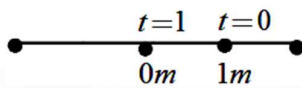
$$2\pi r = 8a$$

$$\frac{r}{a} = \frac{2}{\pi}$$

$$\frac{M_1}{M_2} = \pi \left(\frac{4}{\pi} \right)^2 \times \frac{1}{2} = \frac{8}{\pi}$$

30. At $t = 0, y = 2 \cos \frac{\pi}{3} = 1 \text{ m}$

$$\leftarrow 2A = 4\text{m} \rightarrow$$



$$w = \frac{\pi}{6} = \frac{2\pi}{T} \Rightarrow T = 12 \text{ sec}$$

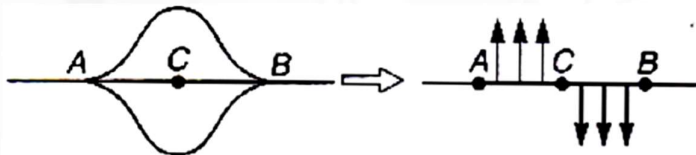
At $t = 1, y = 0$

At $t = 7, y = 0$

Hence from 1 to 7 sec particle travel a distance $= 2A = 4\text{m}$

In time $7 - 1 = 6$ sec particle travels a distance half the length of the oscillation.

31. After two seconds both the pulses will move 4 cm towards each other. So, by their superposition, the resultant displacement at every point will be zero. Therefore, total energy will be purely in the form of kinetic. Half of the particles will be moving upwards and half downwards.

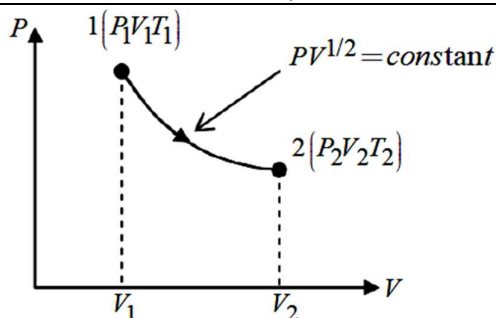


32. $\Delta x = (\mu - 1)t = 1\lambda$

For one maximum shift

$$t = \frac{\lambda}{\mu - 1}$$

33. By theory



34.

From $P-V$ diagram,Given $PV^{1/2} = \text{constant}$

..... (i)

We know that

$$PV = nRT$$

$$P \propto \left(\frac{T}{V}\right)$$

Put in equation

..... (i)

$$\left(\frac{T}{V}\right)(V)^{1/2} = \text{constant}$$

$$T \propto V^{1/2}$$

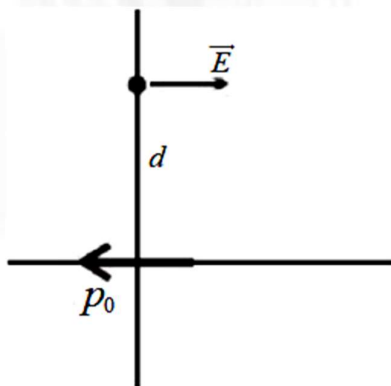
$$\frac{T_2}{T_1} = \sqrt{\frac{V_2}{V_1}}$$

$$\frac{T_2}{T_1} = \sqrt{\frac{2V_1}{V_1}}$$

$$\frac{T_2}{T_1} = \sqrt{2}$$

35.

$$V = 0$$



$$E = \frac{K\vec{P}}{r^3}$$

$$= -\frac{\vec{p}}{4\pi\epsilon_0 d^3}$$

36.

$$\vec{E} = E_0 \hat{n} \sin(\omega t + (6y - 8z)) = E_0 \hat{n} \sin(\omega t + \vec{k} \cdot \vec{r})$$

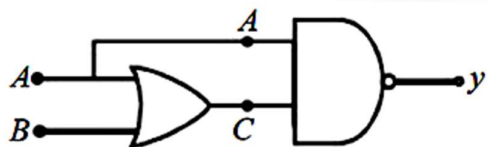


Where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $\vec{k} \cdot \vec{r} = 6y - 8z$

$$\Rightarrow \vec{k} = 6\hat{j} - 8\hat{k}$$

Direction of propagation $\hat{s} = -\hat{k}$

$$= \left(\frac{-3\hat{j} + 4\hat{k}}{5} \right)$$



$$C = A + B$$

$$\text{and } y = \overline{A.C}$$

37.

A	B	$C = (A + B)$	$A.C.$	$y = \overline{A.C}$
0	0	0	0	1
0	1	1	0	1
1	0	1	1	0
1	1	1	1	0

38.

$$L \times \frac{E}{10} \left(-e^{-t/2} \right) \times \frac{-1}{2} = \frac{E}{10} \left(1 - e^{-t/2} \right) \times 10$$

$$e^{-1/2} = 1 - e^{-1/2} ; \quad t = 2 \ln 2$$

39.

$$\Rightarrow T(2\pi r_1 + 2\pi r_2) = h(\pi r_2^2 - \pi r_1^2) \rho g \Rightarrow h = \frac{2T}{(r_2 - r_1) \rho g}$$

40.

The answer will be same for elongation $\Delta \ell = \frac{F\ell}{2AY}$

41.

Mass densities of all nuclei are same so their ratio is 1.

42.

As we know time period of second's pendulum is 2 sec, so Statement-I is incorrect.

Time taken between two extreme points in second's pendulum is 1 sec.

Above Statement is correct because time taken by particle performing SHM between two extreme position is $T/2$.

Here, $T = 2$ sec.

So, time = $2/2 = 1$ sec.

43.

Conceptual

44.

Given phase difference = $\frac{\pi}{4}$ and $\omega = 100 \text{ rad/s}$

$\Rightarrow$ Reactance (X) = Resistance (R)

Now by checking option.

Option (A)

$$R = 1000\Omega \text{ and } X_C = \frac{1}{10^{-6} \times 100} = 10^4\Omega$$

Option (B)

$$R = 10^3\Omega \text{ and } X_L = 10 \times 10^{-3} \times 100 = 1\Omega$$

Option (C)



$$R = 10^3 \Omega \text{ and } X_L = 10^3 \cdot 100 = 10^{-1} \Omega$$

Option (D)

$$R = 10^3 \Omega \text{ and } X_C = \frac{1}{10 \times 10^{-6} \times 100} = 10^3 \Omega$$

45. Height of liquid rise in capillary tube $h = \frac{2T \cos \theta_C}{\rho r g}$

$$\Rightarrow h \propto \frac{1}{r}$$

When radius becomes double height become half

$$\therefore h' = \frac{h}{2}$$

$$\text{Now, } M = \pi r^2 h \times \rho \text{ and } M' = \pi (2r)^2 (h/2) \times \rho = 2M$$

46. $m_C u = (m_B + m_C) v \Rightarrow v = \frac{1 \times 18}{2 + 1} = 6 \text{ ms}^{-1}$

Let t = time after which the string becomes taut.

This will happen when distance moved by A downwards = distance moved by B + C upwards.

$$\text{i.e., } \frac{1}{2} g t^2 = v t - \frac{1}{2} g t^2 \Rightarrow t = \frac{v}{g} = \frac{6}{10} = 0.6 \text{ sec}$$

$$\therefore T = 6$$

47. We know that

$$J = \sigma E$$

$$\Rightarrow J = 5 \times 10^7 \times 10 \times 10^{-3}$$

$$\Rightarrow J = 50 \times 10^4 \text{ A/m}^2$$

Current flowing;

$$I = J \times \pi R^2$$

$$I = 50 \times 10^4 \times \pi (0.5 \times 10^{-3})^2$$

$$I = 5 \times 10^4 \times \pi \times 0.25 \times 10^{-6}$$

$$I = 125 \times 10^{-3} \pi$$

$$X = 5$$

48. Speed of transverse wave is

$$V = \sqrt{\frac{T}{\mu}}$$

$$\ln v = \frac{1}{2} \ln T - \frac{1}{2} \ln \mu$$

$$\frac{\Delta v}{v} = \frac{1}{2} \frac{\Delta T}{T}$$

$$= \frac{1}{2} \times 4$$



$$\frac{\Delta v}{v} = 2\%$$

49. $V_{rms} \sqrt{\frac{3RT_1}{M_0}}$

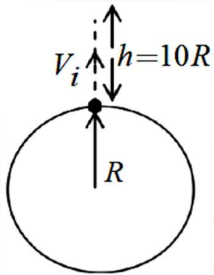
$$200 = \sqrt{\frac{3R \times 300}{M_0}} \quad \dots\dots (1)$$

Also, $\frac{x}{\sqrt{3}} = \sqrt{\frac{3R \times 400}{M_0}} \quad \dots\dots (2)$

$$(1) \div (2)$$

$$\frac{200}{x/\sqrt{3}} = \sqrt{\frac{300}{400}} = \sqrt{\frac{3}{4}}$$

$$\Rightarrow x = 400 \text{ m/s}$$



50.

Here R = radius of the earth

From energy conservation

$$\frac{-Gm_em}{R} + \frac{1}{2}mv_i^2 = \frac{-Gm_em}{11R} + 0$$

$$\frac{1}{2}mv_i^2 = \frac{10}{11} \frac{Gm_em}{R}$$

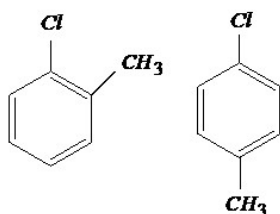
$$V_i = \sqrt{\frac{20}{11} \frac{Gm_e}{R}}$$

$$V_i = \sqrt{\frac{10}{11}} v_e \quad \left\{ \because \text{escape velocity } v_e = \sqrt{\frac{2Gm_e}{R}} \right\}$$

Then the value of x = 10

**CHEMISTRY**

51.

52. CrF_6 is covalent where 'cr' is in state

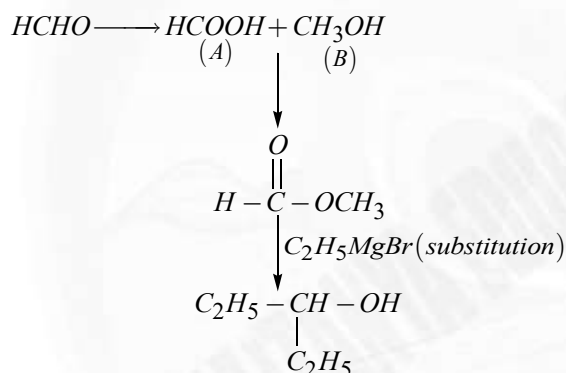
53. NCERT Redox reaction TB Question.

54. $R(r) = 0$ for P orbitals

55. Two stereogenic center

$$\text{i.e. } 2^2 = 4$$

56. Solubility of solid in liquid is unaffected

57. $X = HCHO$ 

58. NCERT Part-I (CBSE) 2022 edition.

59. Millimoles of analyte and titrant are unaffected hang distilled water prior to titration.

60. Chiral carbon of 3-methyl heptan-1-ol remains unaffected

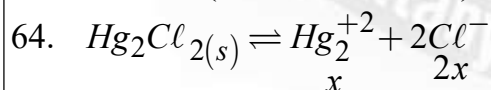
61. P: Effected by ABCD

Q: Effected by ABCD

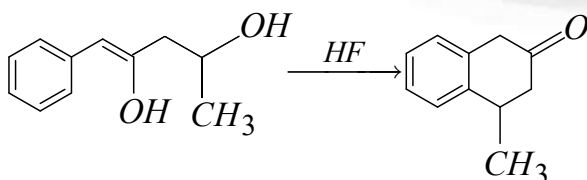
R: Effected by ABCD

S: Effected by AC

62. NCERT (Bimolecule)

63. NCERT (CBSE Part-I 12th)

$$K_{sp} = x(2x^2) = 4x^3$$



65.

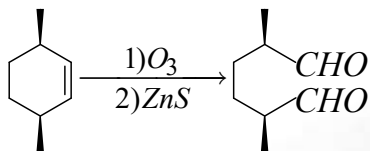
66. NCERT



$$67. \frac{\Delta P}{P_0} = 0.4 = \frac{\Delta_i}{\Delta_i + 3} \Rightarrow 0.6n_i = 1.2$$

$$n_i = 2$$

$$\alpha = 100\%$$



68.

69. XeO_4 is only fluoride and polar.70. Cu^{+2} gives chocolate Brown colour.

$$71. f = e^{-Ea/RT} = \frac{10^{-3}}{100} = 10^{-5}$$

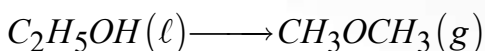
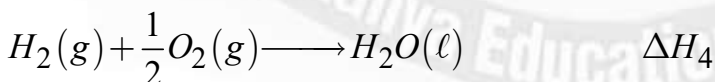
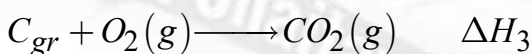
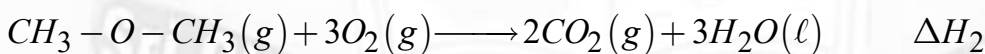
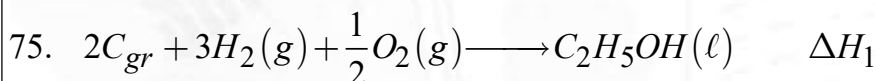
$$K = Ae^{-Ea/RT}$$

$$3 \times 10^{-3} = A \cdot 10^{-5}$$

$$A = \frac{3 \times 10^{-3}}{10^{-5}} = 300 \text{ min}^{-1}$$

72. $|m_l|$ means both $+\ell$ and $-\ell$ values. $4p \rightarrow 2e^\ominus$ $4d \rightarrow 2e^\ominus$ $4f \rightarrow 2e^\ominus$ 

73. All except

74. Paramagnetic: $B_2, O_2^+, C_2^{-1}, He_2^+, O_2$ Fraction BO: O_2^+, C_2^{-1}, He_2^+ 

$$\Delta H = -\Delta H_1 - \Delta H_2 + \Delta H_3 \times 2 + \Delta H_4 \times 3$$

$$= 414 - 188 - 204$$

$$= 22 \text{ kcal/mol}$$